2.1.2 RRL

Title: Benchmarking Landsat-8 Collection 2 Level-2 Land Surface Temperature Accuracy Using SURFRAD Stations: Effects of Seasonality and Atmospheric Water Vapor
Author: Almustafa AbdElkader Ayek ,Mohannad Ali Loho ,Nasser Ibrahem ,Afnan Abdullah Alturki ,Youssef M. Youssef, and Mayada Abdelkader Abdelaziz
https://doi.org/10.3390/atmos17060615

"We present the first comprehensive accuracy assessment using 382 coincident satellite-ground observations collected from seven Surface Radiation Budget Network (SURFRAD) stations distributed across diverse climatic regions of the United States during the period 2023–2025. The validation results indicate strong overall agreement between satellite-derived and ground-measured temperatures, yielding an RMSE of 4.20°C, a coefficient of determination (R^2) of 0.91, and a Pearson correlation coefficient (r) of 0.98. These statistics demonstrate the high reliability of the C2L2 LST product across a wide range of environmental conditions. Nevertheless, a systematic warm bias of 1.75°C was observed, indicating a tendency toward temperature overestimation... Analysis of atmospheric water vapor content revealed significant associations with retrieval errors at high-elevation and arid locations, particularly at FPK (r = 0.78) and DRA (r = 0.75), based on 106 matched observations... Temperature-dependent analyses further demonstrated that retrieval uncertainty increases with surface temperature. Performance progressively deteriorated from cooler to warmer thermal regimes, with RMSE values increasing from approximately 2.05°C for temperatures below 20°C to 5.71°C for temperatures exceeding 40°C."

Key Takeaways from the Extract:

- **Overall Accuracy:** The product demonstrates strong reliability with a coefficient of determination (R^2) of 0.91 and a Pearson correlation coefficient (r) of 0.98.
- **Warm Bias Limitation:** There is a systematic warm bias of 1.75°C, meaning the satellite product tends to overestimate the temperature.
- **High Temperature Limitation:** Retrieval errors increase significantly at higher temperatures, jumping to an RMSE of 5.71°C when surface temperatures exceed 40°C.
- **Atmospheric Water Vapor Limitation:** Atmospheric water vapor content is significantly associated with retrieval errors, especially in arid and high-elevation locations.

Title: Google Earth Engine Open-Source Code for Land Surface Temperature Estimation from the Landsat Series
Author: Sofia L. Ermida, Patrícia Soares, Vasco Mantas, Frank-M. Göttsche, Isabel F. Trigo

https://doi.org/10.3390/rs12091471

(Landsat Data):

- "TOA brightness temperatures for the Landsat's thermal infrared (TIR) channels are provided by the United States Geological Survey (USGS) and are fully available and ready to use in GEE for Landsats 4–8, collection 1. The USGS derives the TOA values from raw orthorectified digital numbers data using the respective calibration coefficients."

- "The data are organized into Tiers based on their quality. Landsat scenes with the highest available data quality are placed into Tier 1, being suitable for time-series analysis, while the remainder are assigned to Tier 2. The results shown in this work are entirely based on Tier 1 data, but the code may equally be applied to Tier 2 data."

- "Cloud coverage information, including cloud shadowing, can be retrieved from the quality assessment band (BQA), which is also made available by the USGS through GEE."

(Processing Chain):

- "Based on this information, the main module, Landsat_LST, loads the respective collections of TOA brightness temperatures (BT) and Surface Reflectance (SR). A cloud mask is applied to both using the quality information bands (module cloudmask)."

Title: Google Earth Engine: Planetary-scale geospatial analysis for everyone
Author: Noel Gorelick, Matt Hancher, Mike Dixon, Simon Ilyushchenko, David Thau , Rebecca Moore
10.1016/j.rse.2017.06.031

"Queries to Earth Engine are based on functional composition and evaluation. Users construct queries by chaining together operations drawn from the Earth Engine library of more than 800 functions, which range in complexity from simple mathematical functions to powerful geostatistical, machine learning, and image processing operations. The library makes it easy to express operations between images using a form of image algebra, and supports higher-order functions: map() and iterate() allow applying arbitrary functions to collections of images, while reduce() is used to compute statistical results in a variety of ways including regional, sliding-window, temporal, spectral and columnar contexts."

"As an illustrative example, a user might wish to compute the difference between two seasonal composites, to highlight the changes due to phenology or snow-cover. A simplistic example of this could be expressed using the Earth Engine client library as the subtraction of two composite images (Listing 1). This code creates two filtered collections, one of all Landsat 8 images for November, December, and January, and a second of all Landsat 8 images from June, July, and August. A temporal median value is computed for each band in each collection (to minimize the effects of cloud and cloud shadows), and the resulting composites are subtracted to compute the change in values. This computational description is represented by the DAG in Fig. 3."

"A temporal median value is computed for each band in each collection **(to minimize the effects of cloud and cloud shadows)**, and the resulting composites are subtracted to compute the change in values."

Title: Google Earth Engine Cloud Computing Platform for Remote Sensing Big Data Applications: A Comprehensive Review
Author: Meisam Amani; Arsalan Ghorbanian; Seyed Ali Ahmadi; Mohammad Kakooei; Armin Moghimi; S. Mohammad Mirmazloumi
doi: 10.1109/JSTARS.2020.3021052

"Remote sensing (RS) systems have been collecting massive volumes of datasets for decades, managing and analyzing of which are not practical using common software packages and desktop computing resources. In this regard, Google has developed a cloud computing platform, called Google Earth Engine (GEE), to effectively address the challenges of big data analysis. In particular, this platform facilitates processing big geo data over large areas and monitoring the environment for long periods of time."

"For instance, according to Hansen et al. [8], it only took 100 h to process 654 178 Landsat-7 images (about 707 terabytes) within GEE and produce a global map of forests. This was reported as a great achievement because if they did not use GEE, this process would have taken a million hours to complete."

Title: Use of normalized difference built-up index in automatically mapping urban areas from TM imagery [PAYWALLED] [ABSTRACT ONLY]
Author: Y. Zha,J. Gao &S. Ni
https://doi.org/10.1080/01431160304987

“Remotely sensed imagery is ideally used to monitor and detect land cover changes that occur frequently in urban and peri-urban areas as a consequence of incessant urbanization. It is a lengthy process to convert satellite imagery into land cover map using the existing methods of manual interpretation and parametric image classification digitally. In this paper we propose a new method based on Normalized Difference Built-up Index (NDBI) to automate the process of mapping built-up areas. It takes advantage of the unique spectral response of built-up areas and other land covers. Built-up areas are effectively mapped through arithmetic manipulation of re-coded Normalized Difference Vegetation Index (NDVI) and NDBI images derived from TM imagery. The devised NDBI method was applied to map urban land in the city of Nanjing, eastern China. The mapped results at an accuracy of 92.6% indicate that it can be used to fulfil the mapping objective reliably. Compared with the maximum likelihood classification method, the proposed NDBI is able to serve as a worthwhile alternative for quickly and objectively mapping built-up areas.”

Title: Estimation of land surface temperature–vegetation abundance relationship for urban heat island studies [PAYWALLED] [ABSTRACT ONLY]
Author: Qihao Weng, Dengsheng Lu, Jacquelyn Schubring
10.1016/j.rse.2003.11.005

“Remote sensing of urban heat islands (UHIs) has traditionally used the Normalized Difference Vegetation Index (NDVI) as the indicator of vegetation abundance to estimate the land surface temperature (LST)–vegetation relationship. This study investigates the applicability of vegetation fraction derived from a spectral mixture model as an alternative indicator of vegetation abundance. This is based on examination of a Landsat Enhanced Thematic Mapper Plus (ETM+) image of Indianapolis City, IN, USA, acquired on June 22, 2002. The transformed ETM+ image was unmixed into three fraction images (green vegetation, dry soil, and shade) with a constrained least-square solution. These fraction images were then used for land cover classification based on a hybrid classification procedure that combined maximum likelihood and decision tree algorithms. Results demonstrate that LST possessed a slightly stronger negative correlation with the unmixed vegetation fraction than with NDVI for all land cover types across the spatial resolution (30 to 960 m). Correlations reached their strongest at the 120-m resolution, which is believed to be the operational scale of LST, NDVI, and vegetation fraction images. Fractal analysis of image texture shows that the complexity of these images increased initially with pixel aggregation and peaked around 120 m, but decreased with further aggregation. The spatial variability of texture in LST was positively correlated with those in NDVI and in vegetation fraction. The interplay between thermal and vegetation dynamics in the context of different land cover types leads to the variations in spectral radiance and texture in LST. These variations are also present in the other imagery, and are responsible for the spatial patterns of urban heat islands. It is suggested that the areal measure of vegetation abundance by unmixed vegetation fraction has a more direct correspondence with the radiative, thermal, and moisture properties of the Earth's surface that determine LST.”

Title: Enhancing the spatial and temporal resolution of satellite-derived land surface temperature in urban environments: A systematic literature review
Author: Robert Andriambololonaharisoamalala , Petra Helmholz ,Ivana Ivánová, Eriita Jones, Susannah Soon , Dimitri Bulatov , Yongze Song
10.1016/j.uclim.2025.102345

On Spatial Resolution Limits:

- *"TIR data usually exhibits a coarser spatial resolution compared to visible and near-infrared (VNIR) data obtained from the same platform because of the difference in energy emission and instrumental limitations (Bisquert et al., 2016; Chen et al., 2016; Ha et al., 2013)."*
- *"Moreover, a trade-off between the swath and the size of the pixel in a sensor limits the ability to provide high spatial and temporal resolutions simultaneously. For instance, Landsat data, the most used product by researchers, offers a relatively high spatial resolution of 30 m; however, its extended revisit time (about 16 days) hinders the potential application."*
- *"Consequently, current moderate resolution TIR data from Earth Observation satellites might not sufficiently capture detailed urban characteristics, thus not suitable for assessing the effectiveness of heat mitigation strategies in cities."*

On the Blending Effect (Mixed Pixels):

- *"For instance, in the case of the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) data, the spatial resolution of VNIR data is 15 m, whereas it is 90 m for the TIR sensors. This coarse resolution may lead to a blending effect in areas where one TIR pixel may cover an area that is bigger than the thermal features within it."*
- *"To study urban climate, better spatial resolution LST data is required to distinguish between urban characteristics such as streets, buildings, and roads."*

Title:
Author: